Supplementary Material for "Decision-Dependent Distributionally Robust Scheduling of Price-Responsive EV Aggregators in Distribution Networks"

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Section, equation, figure and table numbers without the prefix S refer to the main text. References of this document are numbered [S1], [S2], …

S1. Comparison With Representative Studies

Table S1 compares the proposed model with representative studies of EV aggregation, chance-constrained and distributionally robust scheduling, and decision-dependent uncertainty (DDU). It replaces Table 1 of the earlier single-column version.

TABLE S1
Comparison With Representative Studies

Reference	EV flexibility	Network	Exogenous uncertainty	Decision-dependent uncertainty	Uncertainty model
[S1], [S2]	✓	–	✓	–	polytope, joint chance constraint
[S3]	✓	✓	✓	–	GMM joint chance constraint
[S4]	✓	✓	✓	–	moment-based DRO
[S5]	✓	✓	✓	–	DRO joint chance constraint
[S6]–[S8]	✓	–	✓	–	DRO chance constraint
[S9]	✓	✓	✓	–	robust set
[S10]	–	✓	✓	time-of-use price	robust DDU set
[S11]	–	–	✓	production-line choice	robust DDU set
[S12]	✓	–	✓	incentive (station choice)	two-stage stochastic
[S13]	–	–	✓	generic decision	decision-dependent DRO
This paper	✓	✓	✓	incentive price	decision-dependent W_∞ set, exact

S2. Case-Study Data and Parameters

A charging session s with arrival time τ_s^a, plug-in duration d_s, energy e_s and charger rating $\bar{p}_s$ contributes to the connected capacity and to the earliest- and latest-charging energy envelopes:

$$a_{k,t}\overline{P}_k = \textstyle\sum_s \bar{p}_s \mathbb{1}[\tau_s^a \le t < \tau_s^a + d_s] \tag{S1}$$

$$\overline{E}_{k,t} = \textstyle\sum_s \min\{e_s, \eta^{\mathrm{ch}}\bar{p}_s(t-\tau_s^a)^+\Delta t\}, \qquad \underline{E}_{k,t} = \textstyle\sum_s \left(e_s - \eta^{\mathrm{ch}}\bar{p}_s(\tau_s^a + d_s - t)^+\Delta t\right)^+ \tag{S2}$$

where a departed session contributes e_s to both envelopes; these are (16)–(17) of the main text, repeated here for completeness. Sessions are bootstrapped from 300,000 records of the UK domestic charge-point dataset [S14], which preserves the joint distribution of arrival, duration and energy, and the number of sessions per day is Poisson. Two residential, one commercial and one workplace fleet are formed from the same records, following data-driven user typologies [S15]. Photovoltaic errors come from 15-min prosumer data and wind errors from the day-to-day variability of NREL wind speeds [S14]; the repository documents the original sources of all inputs. The degradation cost is linear in discharged energy [S16]. Table S2 in the Appendix of the main text lists all parameters.

S3. Additional Results With Fixed and Co-Optimized Incentives

Table S3 gives the complete fixed-incentive results behind Tables II and III, including the day-ahead cost and the network and reserve failure probabilities separately. With the selected radius, M4 costs 0.8% (33-bus) and 1.4% (123-bus) more than M7; DET exceeds the total cost of M7 by 20% and 40%.

This work was supported by the National Key Research and Development Program of China under Grant 2024YFB4206500.

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TABLE S2
Case-Study Parameters

Item	IEEE 33-bus feeder	IEEE 123-node feeder
Base values	12.66 kV, 10 MVA	4.16 kV, 10 MVA
Peak load	3.715 MW, 2.30 Mvar	3.49 MW (single phase)
Voltage	band 1 ± 0.03 p.u.; $v_0 = 1.0$ p.u.	
Limits	branch 4 MVA; substation 4.32 MVA	branch 6–10 MVA; substation 4.06 MVA
EVA sites	buses 18, 22 residential; 25 commercial; 33 workplace	nodes 114, 113, 85, 111, same types
EVA rating	0.47, 0.46, 0.47, 0.48 MW	0.76, 0.75, 0.76, 0.78 MW
Energy limit	2.90, 2.82, 2.91, 2.80 MWh	4.71, 4.58, 4.72, 4.55 MWh
Departure energy	0.59, 0.58, 0.75, 1.26 MWh	0.96, 0.94, 1.22, 2.04 MWh
Chargers	67, 66, 67, 69 chargers of 7 kW; departure hours 7, 9, 15, 18	
EVA operation	$\eta^{\mathrm{ch}} = \eta^{\mathrm{dis}} = 0.95$; ramp $R_k = 0.3\overline{P}_k$ per hour; $S_k^{\mathrm{cv}} = 1.1\overline{P}_k$	
PV units	buses 18, 33, 14, 30; 0.28, 0.28, 0.28, 0.27 MW	nodes 104, 112, 84, 75; 0.46, 0.46, 0.46, 0.44 MW
Wind turbines	buses 12, 29; 0.45 MW each	nodes 66, 71; 0.73 MW each
Generators	buses 6, 24; 0.25, 0.21 MW	nodes 110, 107; 0.41, 0.34 MW
Generator cost	$a_g = 8,10$ \$/MW^{2}h; $b_g = 150$ \$/MWh	
Costs	$C^{\mathrm{deg}} = 8$ \$/MWh; $C^{\mathrm{sat}} = 500$ \$/MWh; $\rho^{\mathrm{inc}} = 20$ \$/MWh with a fixed incentive and 0, 5, 10 or 20 \$/MWh under co-optimization; 1000 \$/MWh of undelivered reserve	
Price response	model: $\chi_k = 0.5$, $\hat{S}_k^{\mathrm{ch}} = 0.30$, $\hat{S}_k^{\mathrm{dis}} = 0.40$, $\beta = 0.4$, $\pi \in [1, 1.5]$; evaluation: (18) with $\Delta_c = \Delta_d = 0.25$, $m_k \sim \mathcal{U}[0.6, 1.4]$	
Data and risk	300 training, 65 validation and 200 test days; renewable errors scaled by 2.4 and truncated to $[0, C_u]$; $\eta = \tau = 0.05$	
Solver	Gurobi 12.0.3; MIP gap 0.5%; master time limit 120 s (33-bus) and 180 s (123-bus)	

TABLE S3
Out-of-Sample Results With a Fixed Incentive (Complete)

		Cost (\$)		Failure probability (%)			Undelivered	
Method	ε	day-ahead	total	network	reserve	system	reserve (kWh)	p vs M7
IEEE 33-bus feeder								
M1 DET	–	3972.6	4924.1	92.0 [87.4, 95.0]	100.0 [98.1, 100.0]	100.0 [98.1, 100.0]	939.5	<0.001
M2 SAA-CVaR	–	4089.4	4121.9	2.5 [1.1, 5.7]	26.5 [20.9, 33.0]	27.0 [21.3, 33.5]	20.4	<0.001
M3 Moment-DRO	–	4959.1	4971.1	80.0 [73.9, 85.0]	0.0 [0.0, 1.9]	80.0 [73.9, 85.0]	0.0	<0.001
M4 Exo-W$_\infty$	0.050	4124.4	4142.7	0.0 [0.0, 1.9]	9.0 [5.8, 13.8]	9.0 [5.8, 13.8]	6.3	0.454
M5 Robust-DDU	–	4097.7	4100.9	6.0 [3.5, 10.2]	13.5 [9.4, 18.9]	15.0 [10.7, 20.6]	5.3	0.057
M6 DD-CVaR	–	4097.0	4101.3	7.5 [4.6, 12.0]	14.5 [10.3, 20.0]	17.0 [12.4, 22.8]	6.4	0.002
M7 Proposed	0.000	4110.4	4111.0	3.0 [1.4, 6.4]	9.0 [5.8, 13.8]	11.0 [7.4, 16.1]	2.7	–
IEEE 123-bus feeder								
M1 DET	–	3146.0	4690.5	76.0 [69.6, 81.4]	100.0 [98.1, 100.0]	100.0 [98.1, 100.0]	1525.0	<0.001
M2 SAA-CVaR	–	3322.7	3378.5	0.0 [0.0, 1.9]	27.5 [21.8, 34.1]	27.5 [21.8, 34.1]	36.3	<0.001
M3 Moment-DRO	–	4821.9	4841.4	70.0 [63.3, 75.9]	0.5 [0.1, 2.8]	70.5 [63.8, 76.4]	0.0	<0.001
M4 Exo-W$_\infty$	0.050	3376.7	3406.5	0.0 [0.0, 1.9]	9.0 [5.8, 13.8]	9.0 [5.8, 13.8]	10.3	1.000
M5 Robust-DDU	–	3339.3	3344.5	0.5 [0.1, 2.8]	15.0 [10.7, 20.6]	15.0 [10.7, 20.6]	8.7	0.002
M6 DD-CVaR	–	3338.5	3345.4	5.0 [2.7, 9.0]	14.5 [10.3, 20.0]	16.0 [11.6, 21.7]	10.5	<0.001
M7 Proposed	0.000	3358.3	3359.4	0.5 [0.1, 2.8]	9.0 [5.8, 13.8]	9.0 [5.8, 13.8]	4.6	–

The failures of the nominal-response methods M5 and M6 are mainly on the discharging side, 13.5–14.0% against 8.0–9.5% on the charging side. Under co-optimization (Table S4, Fig. S1), M5 and M6 reach a mean price ratio of 1.22 with free incentives, and M7 a mean of 1.05; the maximum price ratio column shows that all decision-dependent models use the upper bound 1.5 in some hours unless the premium is 20 \$/MWh.

TABLE S4
Price Co-Optimization (Complete)

Feeder	Method, ρ^{inc} in \$/MWh	Mean price ratio	Maximum price ratio	Total cost (\$)	System failure (%)
33-bus	M2, price kept at 1	1.000	1.00	4058.2	11.5 [7.8, 16.7]
	M4, price kept at 1	1.000	1.00	4089.5	3.5 [1.7, 7.0]
	M5, $\rho^{\text{inc}} = 0$	1.219	1.50	4049.0	17.0 [12.4, 22.8]
	M6, $\rho^{\text{inc}} = 0$	1.219	1.50	4048.6	18.0 [13.3, 23.9]
	M7, $\rho^{\text{inc}} = 0$	1.052	1.50	4057.3	10.5 [7.0, 15.5]
	M5, $\rho^{\text{inc}} = 5$	1.059	1.50	4057.3	11.5 [7.8, 16.7]
	M6, $\rho^{\text{inc}} = 5$	1.056	1.50	4056.9	13.5 [9.4, 18.9]
	M7, $\rho^{\text{inc}} = 5$	1.012	1.50	4058.4	10.5 [7.0, 15.5]
	M5 to M7, $\rho^{\text{inc}} = 20$	1.002	1.20–1.21	4058.2–4058.7	10.0–11.5
123-bus	M2, price kept at 1	1.000	1.00	3272.5	11.0 [7.4, 16.1]
	M4, price kept at 1	1.000	1.00	3320.1	3.0 [1.4, 6.4]
	M5, $\rho^{\text{inc}} = 5$	1.032	1.50	3272.5	10.0 [6.6, 14.9]
	M6, $\rho^{\text{inc}} = 5$	1.032	1.50	3272.8	12.5 [8.6, 17.8]
	M7, $\rho^{\text{inc}} = 5$	1.012	1.50	3273.6	10.5 [7.0, 15.5]

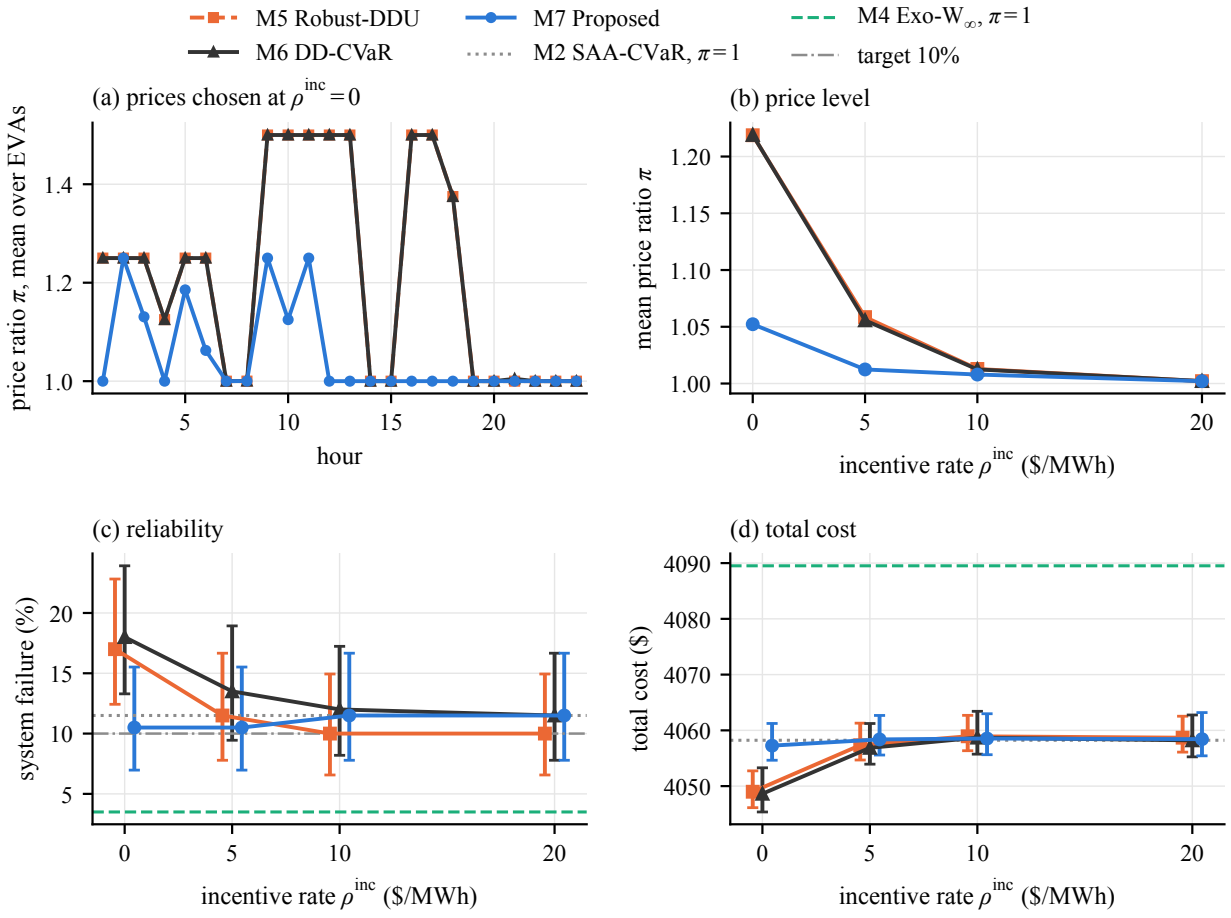


Fig. S1. Price co-optimization on the 33-bus feeder: (a) prices chosen at $\rho^{\text{inc}} = 0$, (b) mean price ratio, (c) system failure and (d) total cost versus the incentive rate.

S4. Additional Sensitivity Results

All results are for the 33-bus feeder with the fixed incentive. Without a sensitivity band, which is M6, 17.0% of the days fail; a band of ±20% brings this to 11.5% for 3.8 \$ per day, and each further 20 points removes about one

more percentage point for about 6 $ per day (Fig. 8a of the main text). With 100 or 50 training days every method degrades, and SAA-CVaR fails on 54% and 72% of the days because its 5% tail then holds only 5 or 2.5 days.

Re-evaluating the stored schedules under true responses with heterogeneity from 0 to ±80% raises the failure rate of M7 only from 10.0% to 13.0%, against 11.5–12.5% to 21.5–23.5% for M5 and M6 and 24.5% to 29.5% for M2. With a linear true response, M7 fails on 10.5% of the days and M6 on 14.0%. With a ±5% voltage band, the network no longer binds and the ranking is unchanged: M7 and M4 fail on 9.0% at 4100.1 and 4129.5 $, against 14.0%, 15.5% and 27.5% for M5, M6 and M2. Raising the degradation cost from 8 to 40 $/MWh under co-optimization cuts the scheduled V2G throughput from 0.64 to 0.05 MWh per day and raises the cost by 10 $, while the failure rates of M7 and M6 remain 10.0% and 12.5%.

S5. Computational Details

Table S5 adds the final optimality gaps to Table VI; the bounds of Algorithm 1 for the proposed method are shown in Fig. 9 of the main text.

TABLE S5
Computational Statistics With a Fixed Incentive (Complete)

	IEEE 33-bus					IEEE 123-bus				
	M2	M4	M5	M6	M7	M2	M4	M5	M6	M7
Iterations	7	6	6	7	6	6	6	6	7	6
Pieces	4287	4601	3988	4036	4125	2513	2881	3511	3491	3335
Time (s)	63	48	14	51	52	41	51	42	90	107
Gap (%)	0.39	0.00	0.00	0.10	0.01	0.00	0.00	0.00	0.00	0.00

References

[S1] M. Zhang, Y. Xu, X. Shi, and Q. Guo, "A fast polytope-based approach for aggregating large-scale electric vehicles in the joint market under uncertainty," IEEE Trans. Smart Grid, vol. 15, no. 1, pp. 701–713, 2024.

[S2] J. Jian, M. Zhang, Y. Xu, W. Tang, and S. He, "An analytical polytope approximation aggregation of electric vehicles considering uncertainty for the day-ahead distribution network dispatching," IEEE Trans. Sustain. Energy, vol. 15, no. 1, pp. 160–172, 2024.

[S3] J. Zhu, Y. Xu, and H. Sun, "Day-ahead joint chance-constrained optimal scheduling of the distribution network with electric vehicle aggregators," IEEE Trans. Smart Grid, vol. 16, no. 6, pp. 5311–5323, 2025.

[S4] X. Shi, Y. Xu, Q. Guo, H. Sun, and X. Zhang, "Day-ahead distributionally robust optimization-based scheduling for distribution systems with electric vehicles," IEEE Trans. Smart Grid, vol. 14, no. 4, pp. 2837–2850, 2023.

[S5] J. Li, Y. Xu, J. Zhang, C. Gao, and H. Sun, "Distributed EV scheduling in distribution networks with reserve market participation under ambiguous probability distribution," Appl. Energy, vol. 383, p. 125269, 2025.

[S6] G. Han and Y.-J. Kim, "Time-efficient and tractable DRCC-based optimal operation of EV charging stations under uncertainty," IEEE Trans. Smart Grid, vol. 17, no. 5, pp. 4469–4481, 2026.

[S7] Y. Li and Z. Li, "Distributionally robust evaluation for real-time flexibility of electric vehicles considering uncertain departure behavior and state-of-charge," IEEE Trans. Smart Grid, vol. 15, no. 4, pp. 4288–4291, 2024.

[S8] M. Bagheri Tookanlou, S. A. Pourmousavi, and M. Marzband, "A three-layer joint distributionally robust chance-constrained framework for optimal day-ahead scheduling of e-mobility ecosystem," Appl. Energy, vol. 331, p. 120402, 2023.

[S9] J. Huang, Y. Wang, C. Gu, L. Cheng, J. Fan, and X. Wang, "DLMP-based congestion management model for power distribution network considering network loss and EV charging demand uncertainty," IEEE Trans. Smart Grid, vol. 17, no. 2, pp. 1414–1429, 2026.

[S10] H. Haghighat, W. Wang, and B. Zeng, "Robust unit commitment with decision-dependent uncertain demand and time-of-use pricing," IEEE Trans. Power Syst., vol. 39, no. 2, pp. 2854–2865, 2024.

[S11] Q. Lv, L. Wang, Z. Li, W. Song, F. Bu, and L. Wang, "Robust optimization for integrated production and energy scheduling in low-carbon factories with captive power plants under decision-dependent uncertainty," Appl. Energy, vol. 379, p. 124827, 2025.

[S12] Y. Gu, Z. Ullah, P. Wang, A. Alferidi, M. Alsolami, B. Lami, and H. M. Hasanien, "Decision-dependent spatiotemporal co-optimization of PV–battery–hydrogen urban EV charging networks under mobility and renewable uncertainty," Appl. Energy, vol. 422, p. 128336, 2026.

[S13] F. Luo and S. Mehrotra, "Distributionally robust optimization with decision dependent ambiguity sets," Optim. Lett., vol. 14, no. 8, pp. 2565–2594, 2020.

[S14] S. Di, "Code and data for decision-dependent distributionally robust scheduling of EV aggregators," GitHub repository, 2026, accessed: Nov. 1, 2026. [Online]. Available: https://github.com/dishiminthu/Distributionally-Robust-Scheduling-of-EVAs-in-Distribution-Networks

[S15] J. R. Helmus, M. H. Lees, and R. van den Hoed, "A data driven typology of electric vehicle user types and charging sessions," Transp. Res. Part C: Emerg. Technol., vol. 115, p. 102637, 2020.

[S16] B. Xu, J. Zhao, T. Zheng, E. Litvinov, and D. S. Kirschen, "Factoring the cycle aging cost of batteries participating in electricity markets," IEEE Trans. Power Syst., vol. 33, no. 2, pp. 2248–2259, 2018.